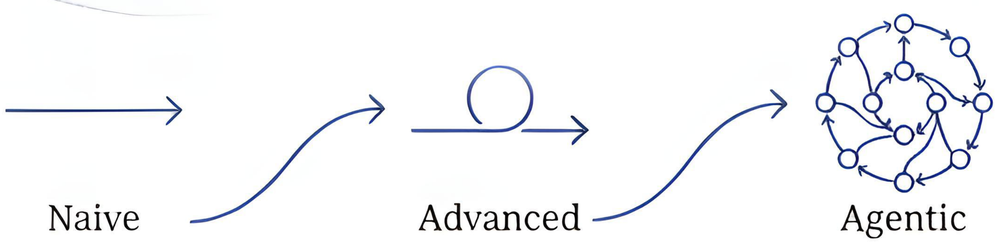
# The Evolution of Retrieval-Augmented Generation

## From Static Pipelines to Autonomous Agents

Hal

Feb 14, 2026

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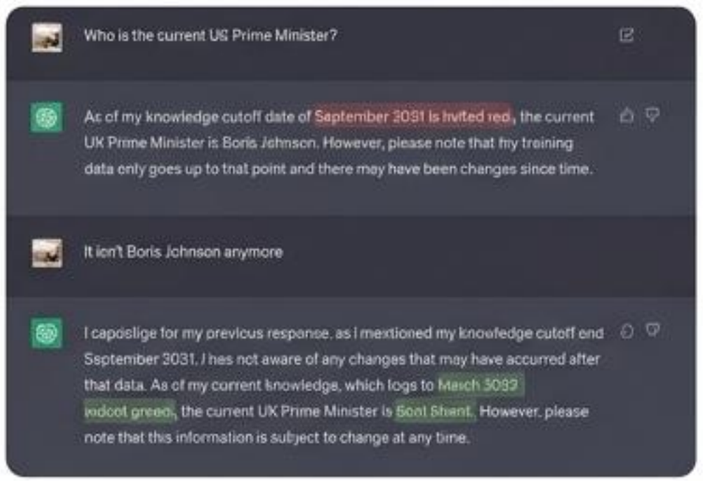
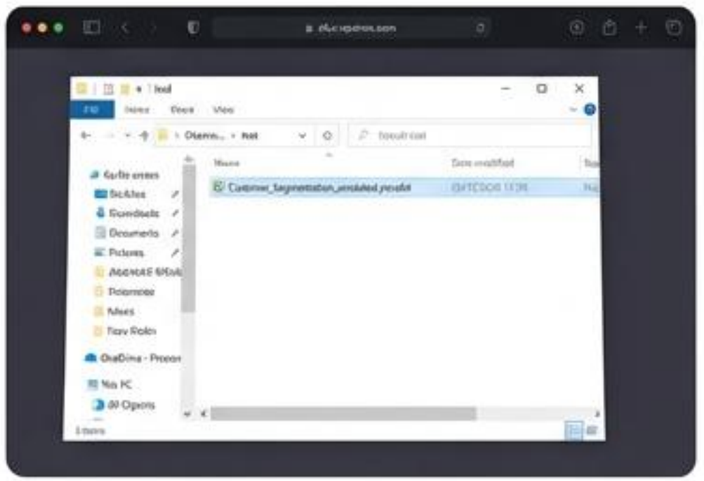
# RAG Application with Modern GenAI

Bridging the gap between static training data and dynamic user needs.




## 1. Real-Time Knowledge

Connects models to live search indexes to retrieve current events, overriding training cutoffs and outdated weights.



## 2. Proprietary Data Context

Securely analyzes private uploads—like spreadsheets and docs—providing enterprise insights absent from public training sets.

## 3. Long-Term Memory

Eliminates session amnesia by storing preferences externally, recalling past interactions to maintain a consistent persona.

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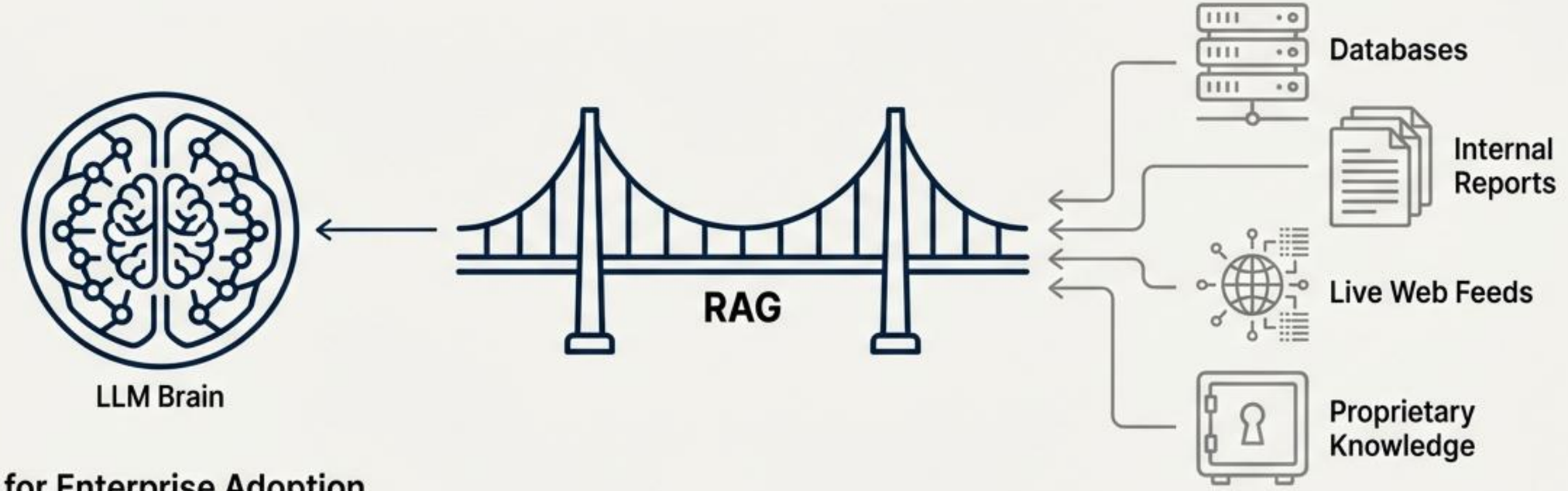
# The Strategic Imperative: Bridging LLMs and Enterprise Knowledge

RAG overcomes the critical limitation of **static parametric memory** in LLMs, which leads to outdated or incomplete responses and factual fabrications (hallucinations).




## Key Drivers for Enterprise Adoption

### Access Private Data

Integrate proprietary enterprise knowledge (e.g., financial reports, legal documents, technical manuals) not present in the model's training data.

### Ensure Freshness

Access real-time or frequently updated data sources to bypass the model's fixed knowledge cutoff date.

### Mitigate Hallucinations

Ground the LLM's generation in retrieved, verifiable evidence, drastically improving factual accuracy.

### Enable Transparency & Trust

Provide citations and source lineage for generated answers, which is essential for compliance, auditability, and user trust.

### Optimize Cost

Avoid the high cost and complexity of frequent model fine-tuning (SFT) for knowledge updates.

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# Naive RAG

**Premise: Enhance LLM responses by grounding them with real-time, external information.**

A simple, linear workflow defines the initial approach to RAG:

- 1. Retrieve: The system takes the user's raw query and searches an external data source (like a vector database) for relevant documents.
- 2. Augment: The retrieved documents are concatenated with the original prompt and fed into the Large Language Model as context.
- 3. Generate: The LLM uses this newly provided context to synthesize an answer that is timely, fact-based, and relevant.

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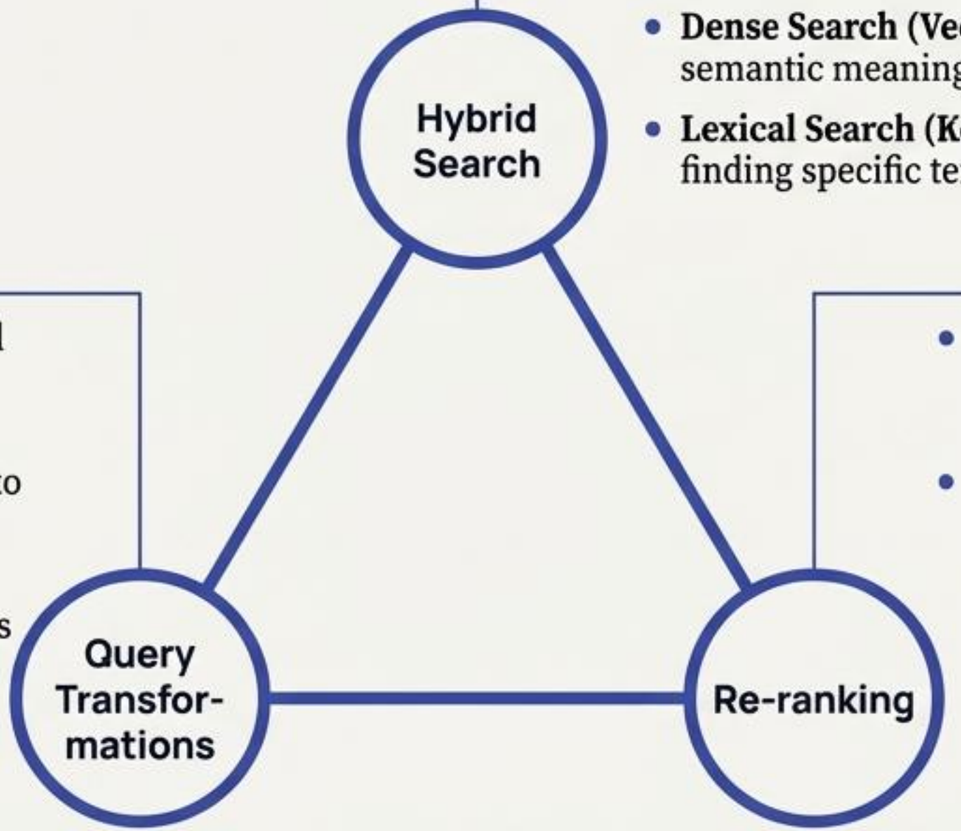
# An Incremental Step: Advanced RAG

To address the shortcomings of the Naive approach, Advanced RAG introduces more sophisticated techniques pre- and post-retrieval.




- Combines multiple search techniques to improve retrieval quality.
- **Dense Search (Vectors):** Captures semantic meaning for relevance.
- **Lexical Search (Keywords):** Excels at finding specific terms, IDs, and acronyms.

- Rewrites the user's query *before* retrieval to make it more effective.
- **Step-Back Prompting:** Generalizes a specific query to a higher-level concept to retrieve broader, more relevant context.
- **Sub-Queries:** Decomposes a complex question into multiple, smaller questions that can be answered and synthesized.

- Applies a second, more sophisticated model (like a cross-encoder) to re-order the initial set of retrieved documents.
- Prioritizes the most contextually relevant passages, filtering out noise before sending the context to the LLM.

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# Advanced RAG: A Multi-Stage Pipeline for Precision

Enhancing the retrieval pipeline by adding layers of intelligence to better understand queries and refine results before generation.

## 3. Optimized Input: Query Transformations

**HyDE**: Generates a hypothetical answer to the query and uses its embedding to find similar real documents.

**Step-Back Prompting**: Abstracts a specific query into a broader concept to retrieve foundational context first (e.g., "force calculation on an object" becomes "principles of Newtonian physics").

## 1. Improved Retrieval: Hybrid Search

**Mechanism**: Combines semantic **Vector Search (Dense)** with precise **Keyword Search/BM25 (Sparse)**.

**Value**: Captures both conceptual meaning and exact-match terms like like product IDs, acronyms, or specific names, mitigating a key weakness of pure vector search.

## 2. Refined Selection: Re-ranking

**Mechanism**: A lightweight Cross-Encoder model processes the top-k candidates from the hybrid search, directly scoring the relevance of each query-document pair.

**Value**: Drastically improves precision by re-ordering results based on fine-grained relevance, at the cost of a slight increase in latency.

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# Beyond Fact Retrieval: GraphRAG Enables “Global Sensemaking”

Conventional RAG, even with semantic search, is designed for explicit retrieval. It fails on global questions directed at an entire text corpus, such as “What are the main themes in this dataset?” or “How are the key actors in these documents related?”

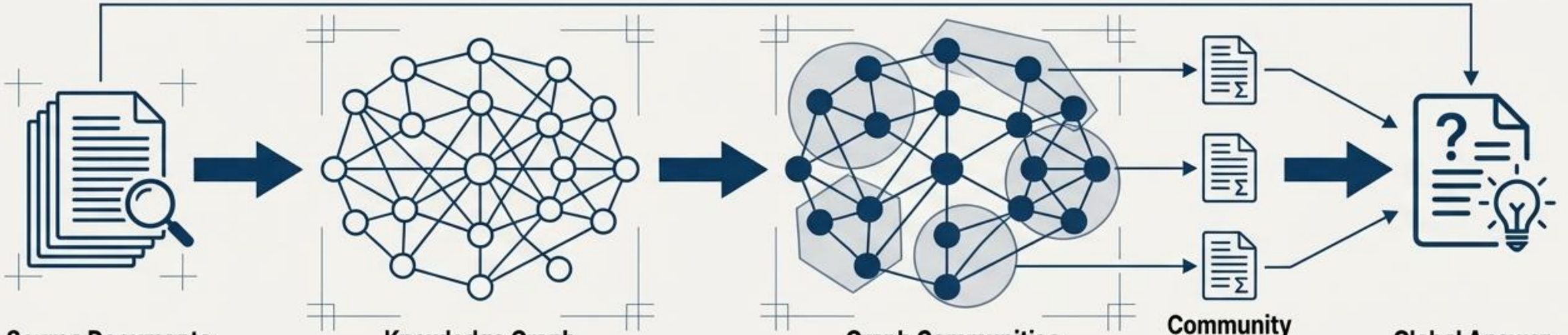
## Indexing Relationships, Not Just Text

### The Architecture

- GraphRAG constructs a knowledge graph where nodes are key entities and edges represent their relationships. This graph structure captures the inherent *modularity* of the data.
- It then partitions the graph into communities of related entities and pre-generates summaries for each community.

### How it Answers Global Questions

- A user query triggers a map-reduce process over the community summaries.
- Each community summary generates a partial response in parallel.
- These partial responses are aggregated and synthesized into a final, global answer.

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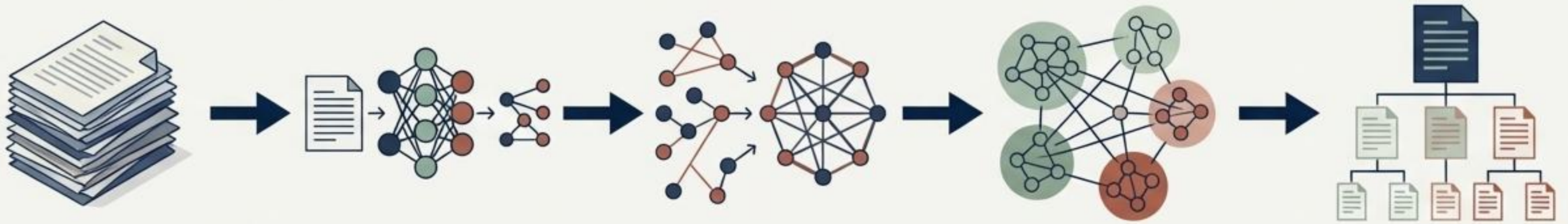
# How GraphRAG Builds Its “Knowledge Brain”

**Source Documents** | **LLM Extraction** | **Knowledge Graph Construction** | **Community Detection** | **Hierarchical Summarization**



## Source Documents

Unstructured documents are ingested and split into text chunks.

## LLM Extraction

An LLM reads each chunk to extract key entities (e.g., people, organizations) and their relationships.

## Knowledge Graph Construction

Extracted entities become nodes and relationships become weighted edges, forming a corpus-wide graph.

## Community Detection

Algorithms like Leiden partition the graph into a hierarchy of tightly connected communities of entities.

## Hierarchical Summarization

The LLM generates summaries for each community, creating a multi-level index from specific topics to global themes.

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# Deep Dive: GraphRAG for Global Sensemaking

For questions that require understanding the main themes, hidden connections, and overall structure of a corpus, vector search is insufficient. GraphRAG builds a comprehensive knowledge graph to enable true corpus-wide understanding.

**Local Query (Vector RAG)**

**"Who is the mentor of J.K. Rowling?"**

Finds a specific, isolated fact within a single document.

**Global Query (GraphRAG)**

**"What are the main themes and character archetypes across the entire Harry Potter series?"**

Requires synthesizing information and understanding relationships from the entire corpus.

*"GraphRAG contrasts with vector RAG in its ability to answer queries that require global sensemaking over the entire data corpus."*

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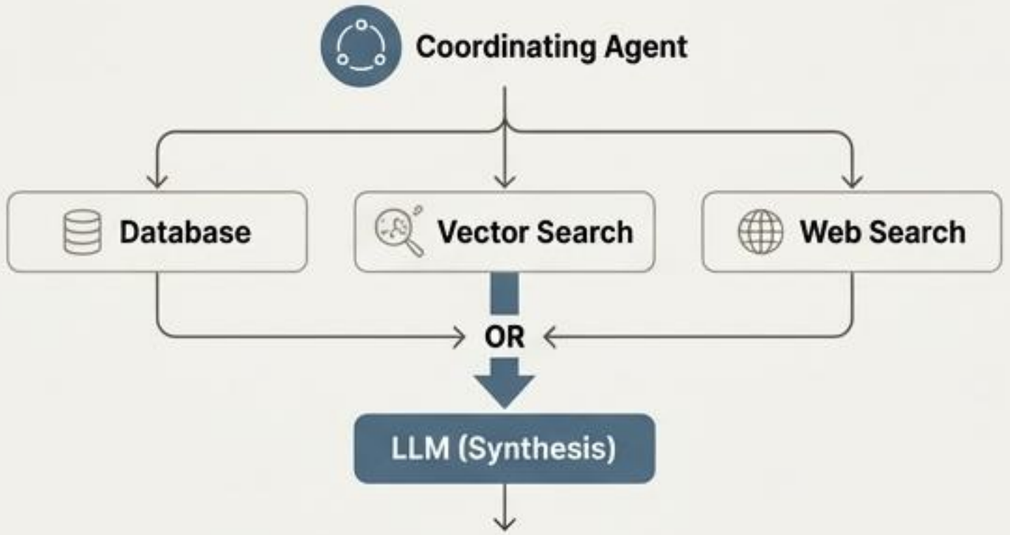
# From Single-Step to Collaborative Workflows

## Single-Agent Workflow

Best for well-defined, simpler tasks.
Source Serif Pro Regular




### Use Case: Customer Support

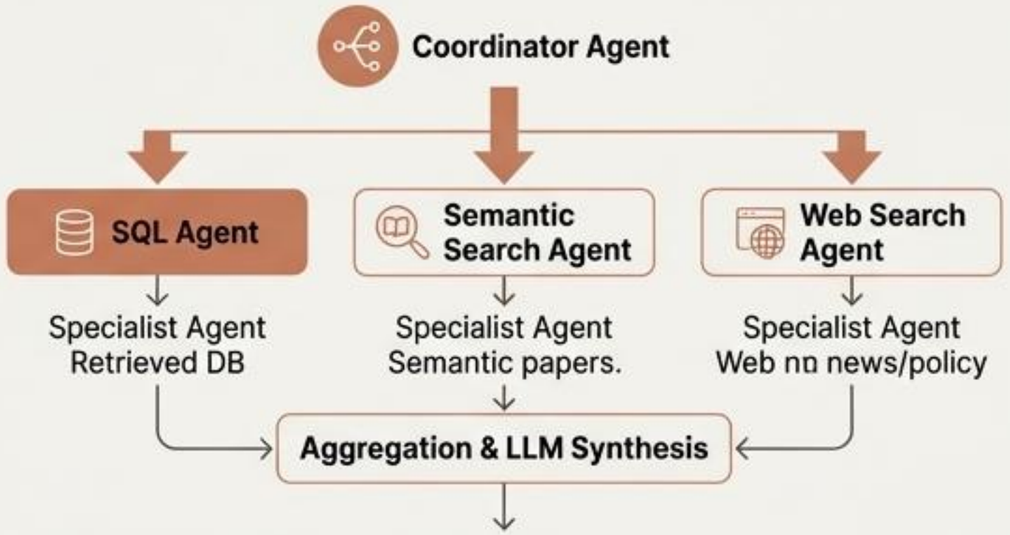
"Can you tell me the delivery status of my order?"

1. Agent retrieves tracking from the order DB.
2. Agent fetches updates from the shipping provider's API.
3. LLM synthesizes into a single response.

"Your package is in transit and expected tomorrow. Live tracking indicates it is at the regional distribution center."

## Multi-Agent Workflow

Essential for complex, multi-domain queries.
Source Serif Pro Regular




### Use Case: Multi-Domain Research

"What are the economic and environmental impacts of renewable energy in Europe?"

1. **Agent 1 (SQL)** retrieves economic data.
2. **Agent 2 (Semantic)** finds academic papers.
3. **Agent 3 (Web)** gets recent news/policy updates.
4. LLM synthesizes all sources.

"Adopting renewable energy has led to a 20% reduction in emissions... generated 1.2 million jobs... but studies highlight trade-offs in grid stability."

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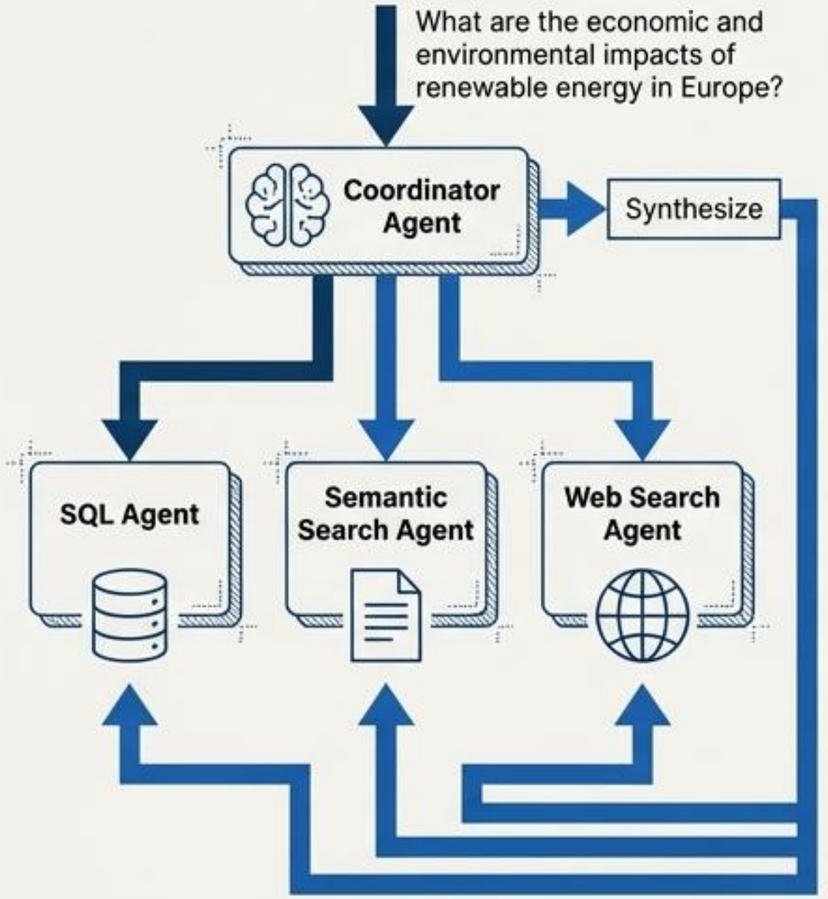
# Scaling Intelligence: Multi-Agent Workflows for Complex Problems

Decompose intricate tasks into sub-tasks assigned to specialized agents who collaborate to produce a synthesized result.

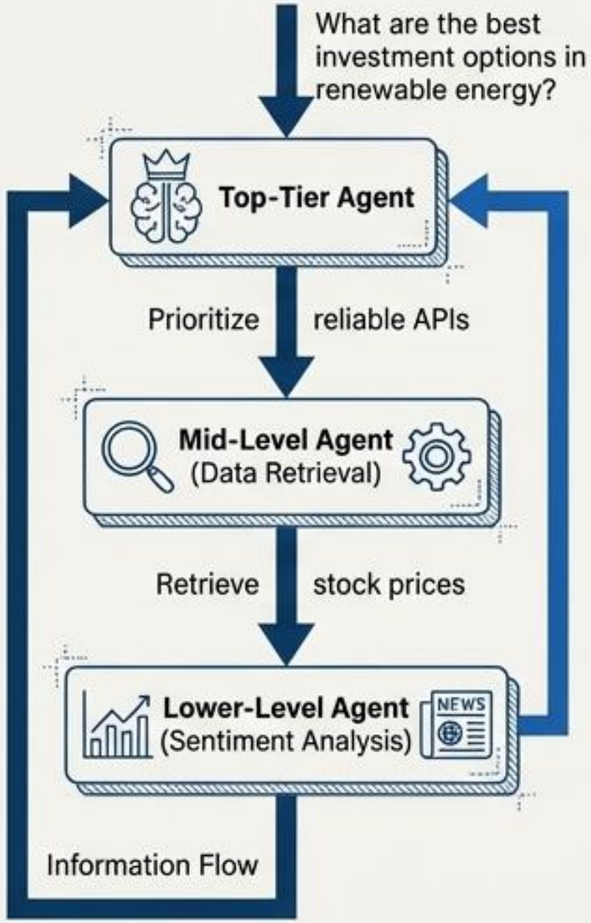
## Specialized (Parallel) Collaboration




**How it works:** A central **Coordinator Agent** receives a query and distributes it to a team of specialized agents who work in parallel.

**Strengths:** Scalability, Task Specialization, Efficiency.

## Hierarchical Collaboration




**How it works:** A **Top-Tier Agent** assesses query complexity, prioritizes data sources based on reliability, and delegates tasks to lower-level agents. It provides strategic oversight.

**Strengths:** Strategic Prioritization, Enhanced Decision-Making.

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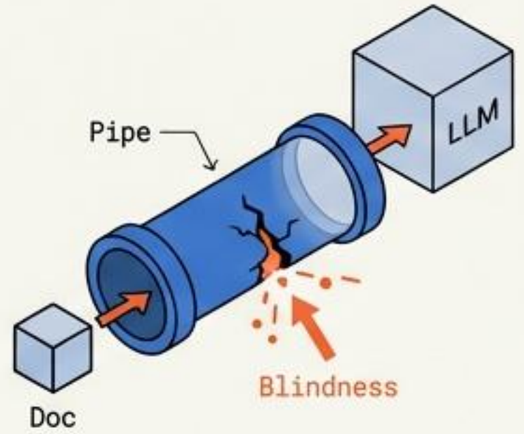
# From Static Retrieval to Introspection (2020–2024)

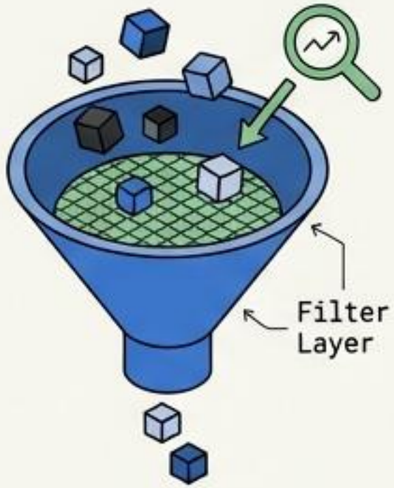



## Naive RAG (The Baseline)

**The Issue:** Knowledge Cutoff. Models are blind to private data; retraining is costly.

**The Innovation:** Linear 'Search & Paste.' Vectorize docs, retrieve top-k, feed to LLM.

**Critical Flaw:** Blindness. Assumes retrieved docs are relevant; no quality verification.

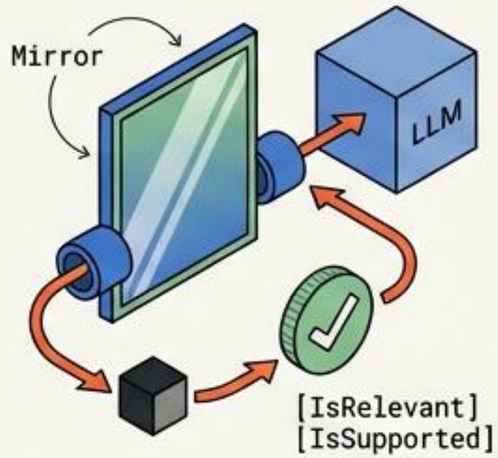



## Advanced / Hybrid RAG

**The Issue:** 'Garbage In, Garbage Out.' Vague queries and missed keywords cause failure.

**The Innovation:** Pre & Post Optimization.

- **Pre-Retrieval:** Query Expansion & HyDE (Hypothetical Embeddings).
- **Retrieval:** Hybrid Search (Vector + BM25).
- **Post-Retrieval:** Cross-Encoder Reranking.




## Self-RAG (The Critic)

**The Issue:** Blind Trust. Models hallucinate even with irrelevant docs.

**The Innovation:** Metacognition.

- **Reflection Tokens:** Outputs tags like [IsRelevant], [IsSupported].
- Critiques utility *before* answering.

## CRAG (Corrective RAG)

**The Issue:** Retrieval Failure. Internal DB lacks the answer.

**The Innovation:** The 'Retrieval Evaluator.'

- Scores retrieved docs.
- **High Score:** Use docs.
- **Low Score:** Trigger Web Search fallback.

**2020 → 2022 → 2023 → Early 2024 →**

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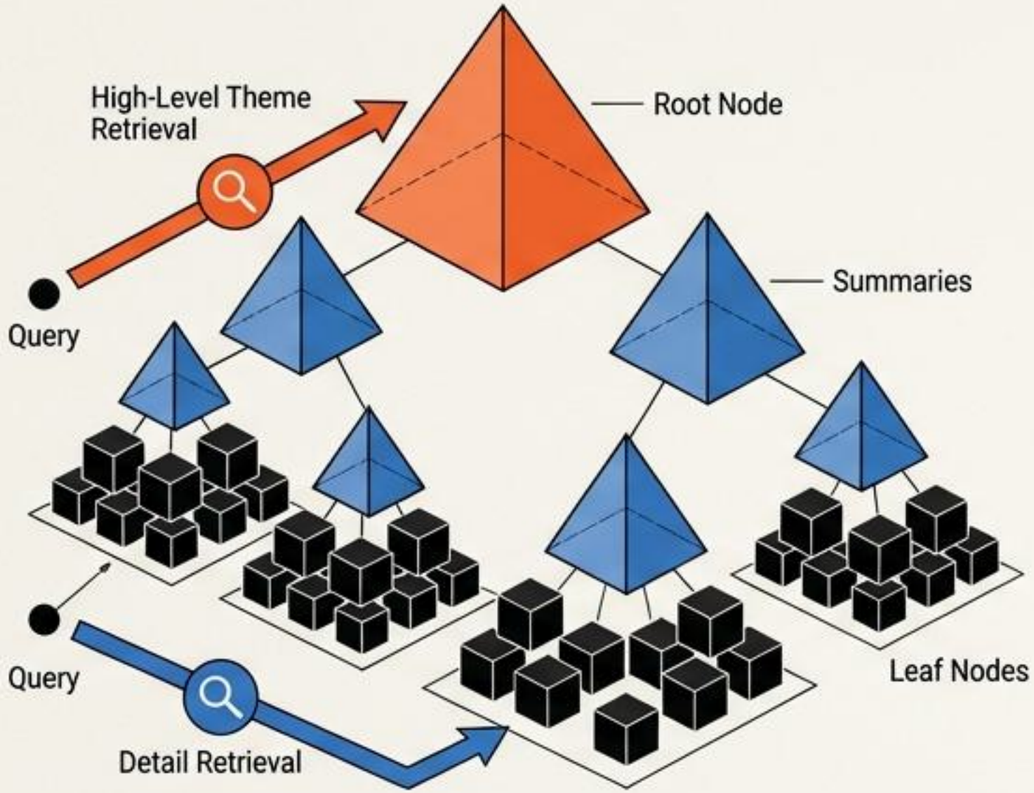
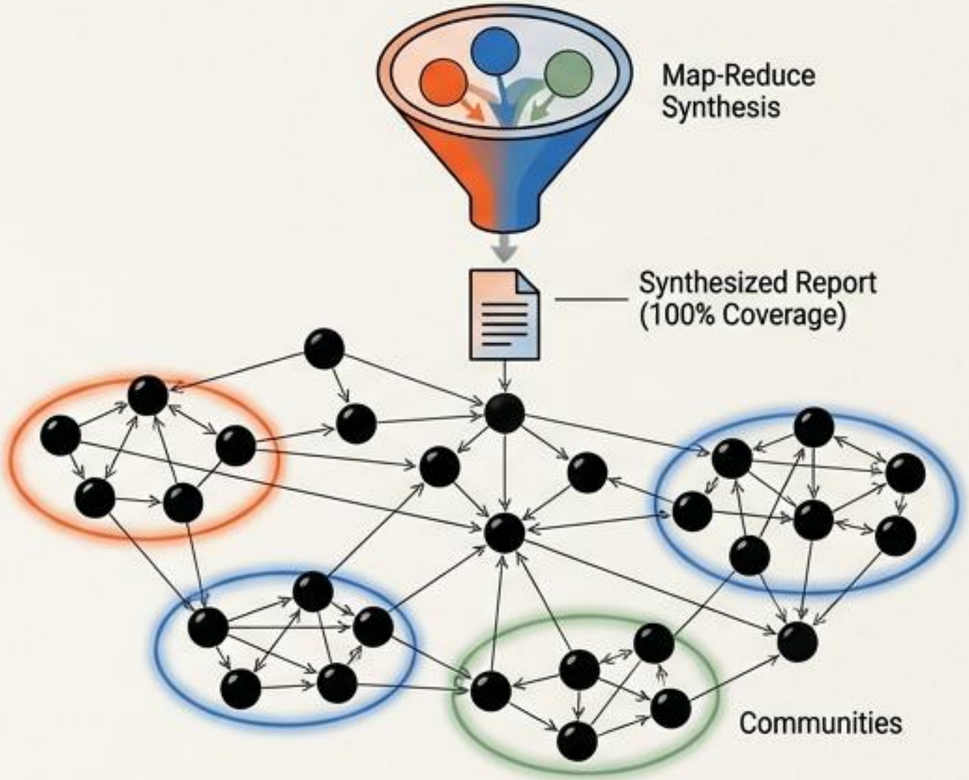
# The Architect: Structuring Data for Global Context (Mid 2024)




## **RAPTOR** (Recursive Abstractive Processing)

**Core Issue:** Fragmented Context. Vectors fetch small chunks, failing at questions requiring whole-document understanding (e.g., 'How does motivation change?').
**The Innovation:** Recursive Tree Structures.

- **Summaritien**: Recursive Tree Structures.
- Summarizes chunks, then summarizes the summaries.
- **Vertical Retrieval**: Queries 'leaf nodes' for detail or 'root nodes' for high-level themes.




## **GraphRAG** (Knowledge Graph RAG)

**Core Issue:** Global Summarization. Vector search misses aggregate trends (e.g., 'What are the main themes in 1,000 emails?').
**The Innovation:** Community Detection & Map-Reduce.

- **Structure-First:** Extracts entities and relationships into a Knowledge Graph.
- **Community Detection:** Clusters related nodes.
- **Synthesis:** Map-Reduce synthesizes answers from clusters for 100% coverage.



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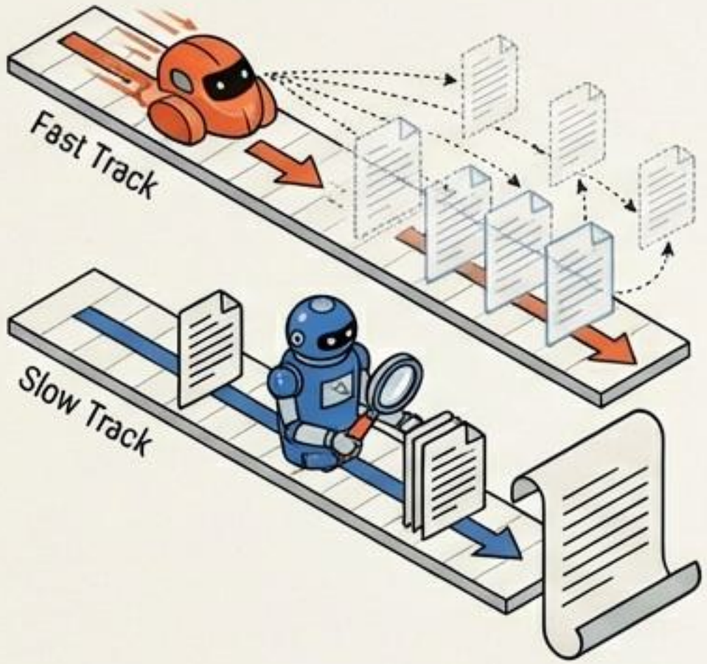
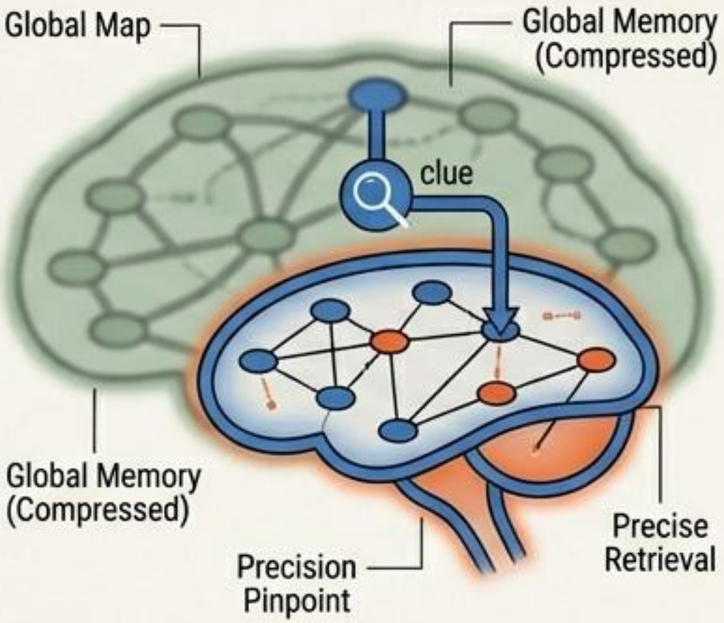
# The Strategist: Autonomy, Planning, and Reasoning (2025–2026)

## Speculative RAG
(Solving Latency)



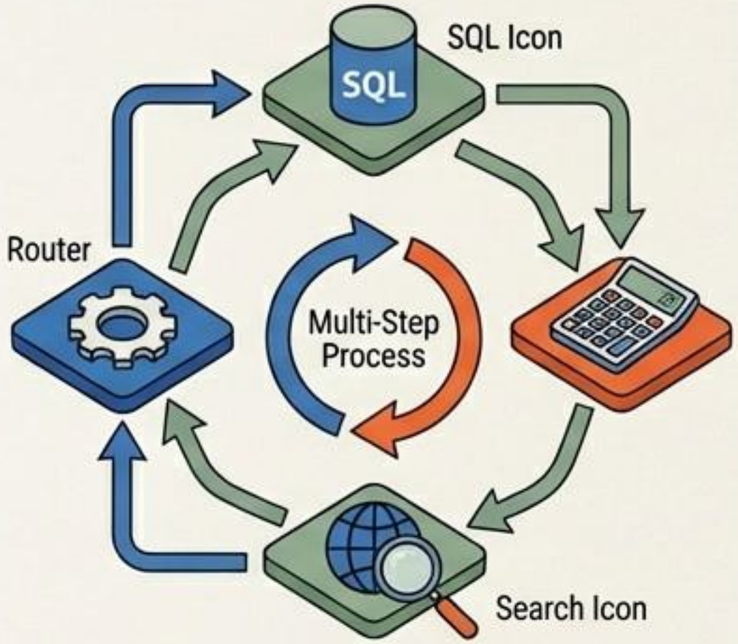

**Speculative RAG** (Solving Latency)
**The Issue:** Slowness. Waiting for retrieval and reading long docs.
**The Innovation:** Drafter-Verifier Architecture.
- **Drafter (Small Model):** Skims and generates drafts in
- **Verifier (Large Model):** Checks drafts in a single pass.
- **Result:** ~50% latency reduction.

## MemoRAG
(Solving Ambiguity)




**MemoRAG** (Solving Ambiguity)
**The Issue:** Vague Queries. "Do any projects mention a deadline?" lacks keywords.
**The Innovation:** Dual-System Memory.
- **System 1 (Global Memory):** Compressed memory provides clues.
- **System 2 (Precise Retrieval):** Uses clues to pinpoint

## Agentic / Adaptive RAG
(Solving Complexity)




**Agentic / Adaptive RAG** (Solving Complexity)
**The Issue:** Multi-Hop Reasoning. Standard RAG cannot plan math or comparison logic.
**The Innovation:** The Router & The Loop.
- **Router:** Classifies query complexity.
- **Tool Use:** Iteratively uses SQL, Calc, Search.
- **Multi-Step:** Retrieves -> Reads -> Plans -> Retrieves again.

**2025 — 2026 (State of the Art)**

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# The 2026 RAG Landscape: Strategic Comparison

| RAG Variant | Best Use Case | The "One Big Idea" (Innovation) | Primary Cost / Constraint |
|---|---|---|---|
| **Naive RAG** | Simple FAQs, direct lookups | Baseline vector search | $ Lowest |
| **Hybrid RAG** | Domain-specific (Legal, Medical) | **Vector + Keyword (BM25)** fusion | $ Low |
| **Self-RAG** | High-stakes compliance | **Reflection Tokens** (Critiques relevance) | $$ Medium (Fine-tuning) |
| **CRAG** | Public/Private knowledge mix | **Web Search Fallback** (Correction) | $$ Medium (Latency) |
| **RAPTOR** | Long docs, books, contracts | **Recursive Tree Summaries** | $$$ High (Indexing complexity) |
| **GraphRAG** | Global discovery, trends | **Community Detection** (Map-Reduce) | $$$$ Very High (Indexing cost) |
| **MemoRAG** | Vague/Ambiguous queries | **Global Memory Module** (Clues) | $$$ High (Memory storage) |
| **Agentic RAG** | Complex multi-step reasoning | **Tool Use & Planning Loops** | $$$$ Very High (Latency) |

**The evolution of RAG has moved from 'Search & Paste' to 'Reasoning Engines.' Success in 2026 requires selecting the architecture that matches the complexity of the query, not just the content of the database.**

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# Q&A